



Clothes Donation System

For

WebX CA Project

**Third Year (Semester-VI) Bachelors in Information Technology, Autonomous
Program**

By

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Class: D15A

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_Department of Information Technology

**Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai**

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Vivekanand Education Society's Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)
NAAC accredited with 'A' grade

Certificate

This is to certify that project entitled

”Clothes Donation System”

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In fulfilment of degree of BE. (Sem. IV, VI ,VII) in Information Technology for WebX CA
Project is approved.

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Chapter 1

Introduction

1.1 Overview

The Clothes Donation App is a full-stack web-based platform designed to simplify and digitalize the process of donating clothes. Built using React.js for the frontend, Flask for the backend, and MongoDB as the database, the app facilitates seamless interactions between individual donors and Non-Governmental Organizations (NGOs). It acts as a digital bridge that connects donors willing to contribute clothing items with NGOs in need of resources, enhancing efficiency, transparency, and accessibility in the donation process.

Users can submit donation requests by specifying the type and quantity of clothes, along with pickup location details. These requests are then reviewed by NGOs, who can choose to accept or reject them based on their current needs and logistical capacity. Once accepted, the donation is scheduled for pickup, and the donor is kept informed of the status throughout the process. The platform also maintains a donation history section for each user, allowing them to track and review their past contributions.

The application supports the values of community service and sustainability by encouraging reuse and responsible giving, while also ensuring that NGOs can manage their resources effectively.

Github Link: <https://github.com/madhurajangale/Clothes-Donation-System>

1.2 Objectives

The primary objectives of the Clothes Donation App are:

- **To simplify the process of donating clothes** by providing a user-friendly, digital interface for donors and NGOs.
- **To establish a transparent workflow** for donations, where users can view real-time statuses such as pending, accepted, or picked up.
- **To empower NGOs** by giving them control over the acceptance and scheduling of donations based on their capacity and availability.
- **To promote sustainability** through the reuse of clothing items, reducing textile waste and supporting underprivileged communities.
- **To encourage repeated participation** by providing users with donation histories and positive feedback loops.
- **To ensure secure data handling** and smooth communication between users and NGOs using reliable backend technologies and database management.

Chapter 2

2.1 Literature Survey

1. G. Pandey and A. Kumar, "An Examination on Food, Clothes and Books Donation Based Android Application," 2022 Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), Mandya, India, 2022, pp. 1-6, doi: 10.1109/ICERECT56837.2022.10059757.

This paper discusses an Android-based platform for organizing and managing donations of food, clothes, and books. It focuses on ensuring accessibility and ease of use for both donors and NGOs, stressing the importance of categorization and user tracking mechanisms. It laid the groundwork for multi-category donation systems.[1]

2.N. V. Harika, G. L. Pravalika, M. A. Kumar and A. Jashuva, "Streamlining Donations to Orphanages and Senior Care: A Necessary Step," 2024 International Conference on Sustainable Communication Networks and Application (ICSCNA), Theni, India, 2024, pp. 1539-1544, doi: 10.1109/ICSCNA63714.2024.10864148.

This work highlights how digital tools can streamline donations to sensitive institutions like orphanages and elderly care homes. The system includes donor verification, tracking of donation history, and scheduling logistics — many of which are applicable to clothing donation workflows.[2]

3.C. Varghese, D. Pathak and A. S. Varde, "SeVa: A Food Donation App for Smart Living," 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC), NV, USA, 2021, pp. 0408-0413, doi: 10.1109/CCWC51732.2021.9375945.

This paper introduces an IoT-enabled smart donation platform focused on food redistribution. Although centered on food, SeVa's architecture emphasizes intelligent scheduling, feedback loops, and real-time tracking — components also relevant in a clothes donation system. It integrates AI and HCI principles to ensure intuitive user experience and efficient logistics.[3]

Chapter 3

3.1 Project definition

The **Clothes Donation App** is a full-stack web application developed using **React.js** for the frontend, **Flask** for the backend, and **MongoDB** as the database. The app serves as a digital bridge between individual donors and NGOs, making the process of donating clothes simple and efficient.

Users can fill out a **donation form** for a specific NGO, providing details such as clothing type, quantity, and pickup location. Once submitted, the request is sent to the NGO, which can **accept or reject** the donation based on their availability and capacity. Accepted donations are then scheduled for pickup.

The app also features a **donation history** section where users can track their past donations and view statuses such as pending, accepted, or picked up. This helps ensure transparency and encourages continued participation.

By streamlining the donation workflow, the Clothes Donation App promotes community support and sustainability through an easy-to-use online platform.

3.2 Technology Stack

Frontend

- **React.js** – JavaScript library for building responsive user interfaces.
- **CSS** – For custom styling and layout design.
- **Axios** – For making HTTP requests to the backend API.

Backend

- **Flask** – Lightweight Python web framework used to handle APIs and server-side logic.
- **Flask-CORS** – To manage cross-origin requests between frontend and backend.
- **Flask-Mail** – For implementing email notifications to users/NGOs (e.g., donation request confirmation, acceptance alerts).

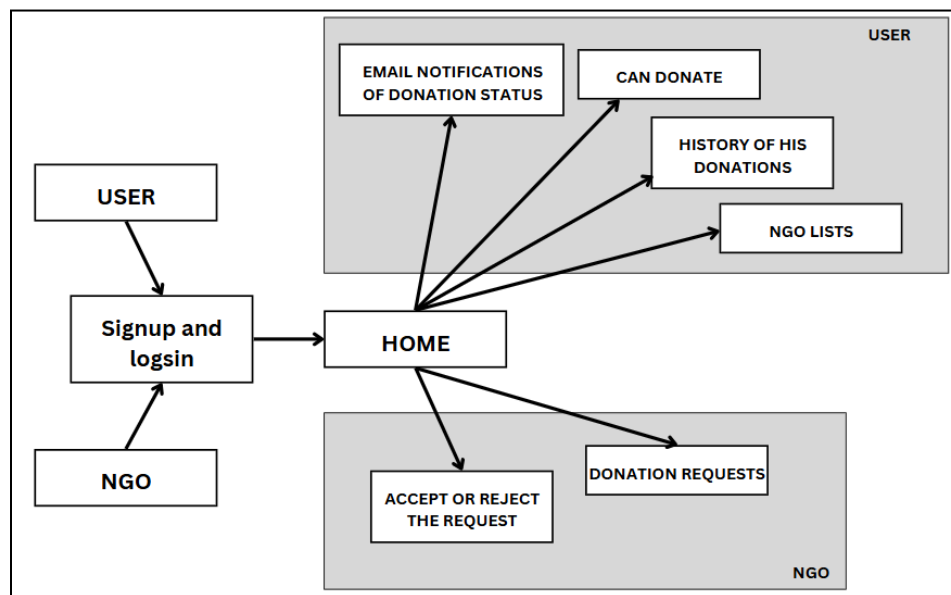
Database

- **MongoDB (MongoDB Atlas)** – NoSQL database to store user data, donation records, and NGO profiles.

3.3 Features

- User Donation Form**
 Users can fill out a form to donate clothes to a selected NGO, specifying item details and pickup location.
- NGO Request Management**
 NGOs receive donation requests, review and accept or reject based on their capacity.
- Donation History Tracking**
 Users can view a detailed history of all their donations along with status updates (Pending, Accepted, Picked Up).
- Email Notifications**
 Donors receive automated emails using Flask-Mail upon request submission and NGO response (acceptance/rejection).
- User-Friendly Interface**
 Clean and responsive design built with React and CSS, offering smooth navigation and an intuitive experience.

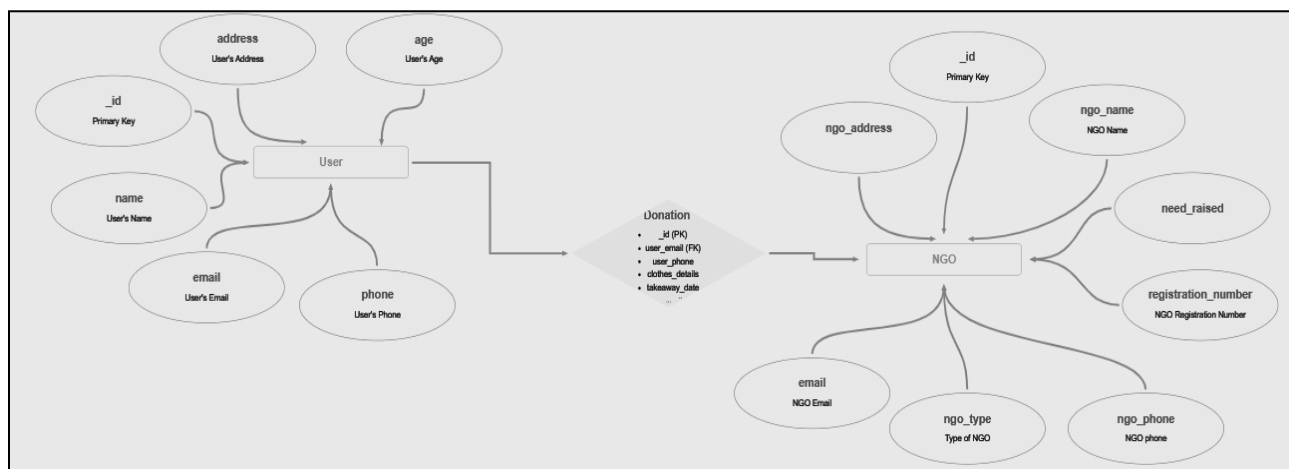
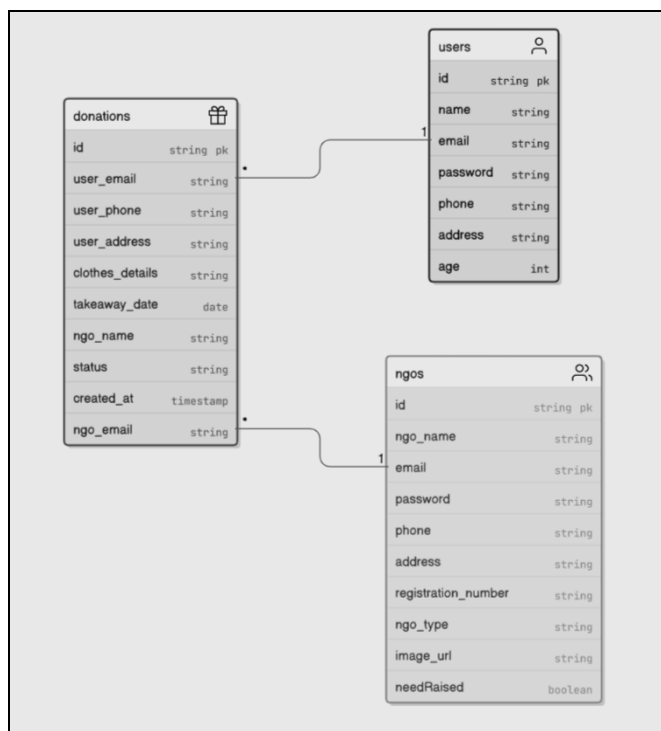
3.4 Flow Diagram



URL REPRESENTATION-

— /login	→ Login page
— /signup	→ Signup page
— /home	→ User's homepage (after login)
— /ngoreq	→ NGO dashboard to see/manage donation requests
— /ngos	→ List of available NGOs for users to donate to
— /history	→ Donation history (user view)
— /profile	→ Profile page (user or NGO)

3.5 ER Diagram



3.6 Hardware Requirements

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8GB (Recommended: 16GB)
- **Storage:** At least 10GB free disk space
- **Operating System:** Windows 10/11, macOS, or Linux

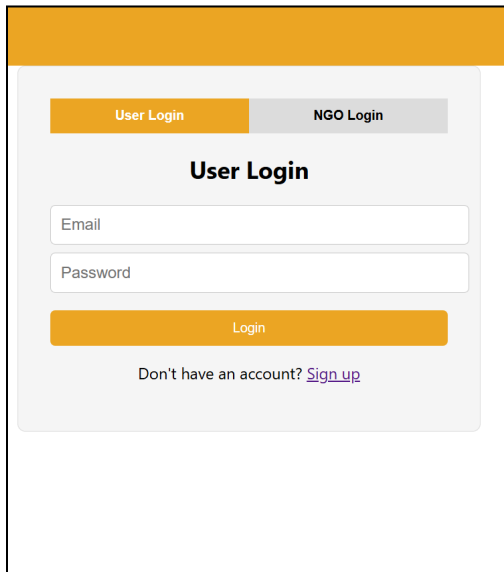
3.7 Software Requirements

- **Node.js** (v18.16.1 or later) – Required for React frontend
- **npm** (v9.5.1 or later) – For managing frontend dependencies
- **Python** (3.11.9 or later) – Required for Flask backend
- **MongoDB Atlas** – Cloud-based NoSQL database
- **VS Code** – Recommended IDE for development
- **Postman** – For testing backend APIs

Chapter 4

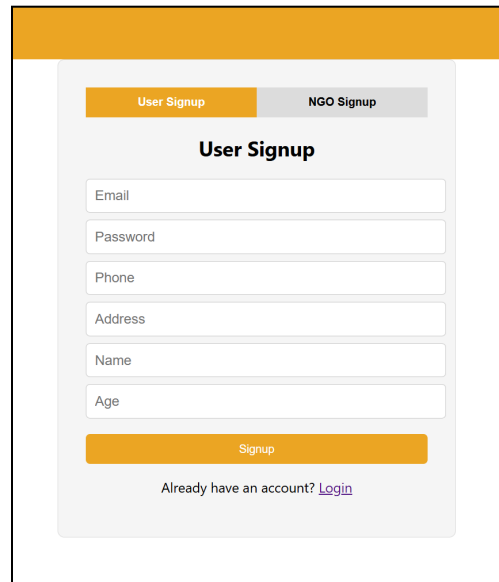
4.1 Output (frontend and Backend)

Login page-



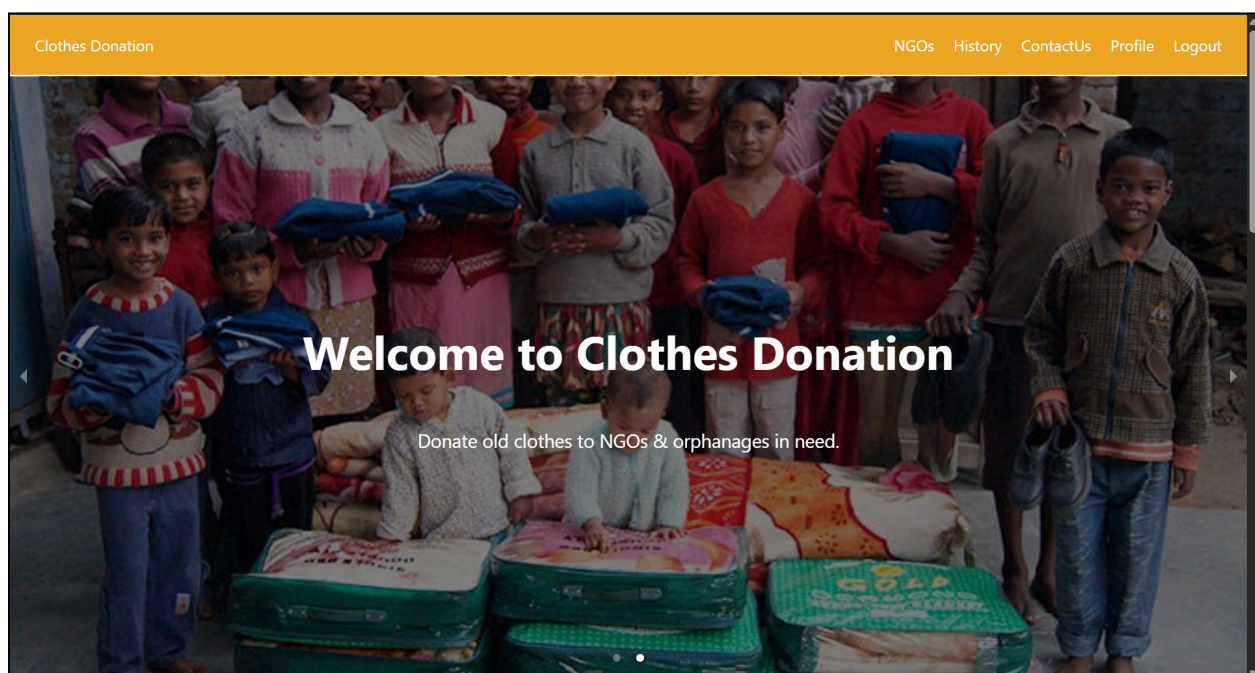
The login page features a yellow header bar. Below it, there are two tabs: 'User Login' (active) and 'NGO Login'. The 'User Login' section contains a title 'User Login', an email input field, a password input field, and a yellow 'Login' button. At the bottom, there is a link: 'Don't have an account? [Sign up](#)'.

Signup page-

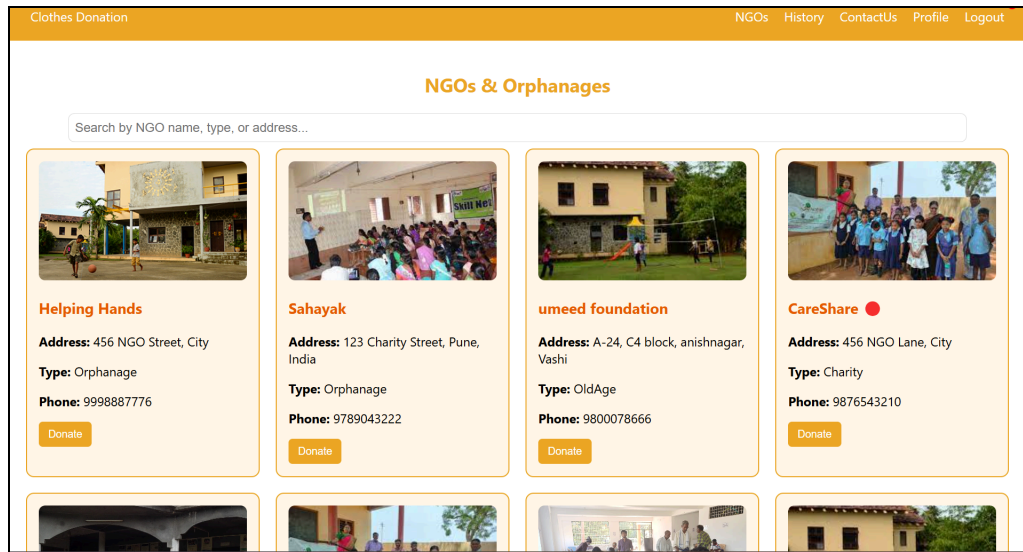


The signup page features a yellow header bar. Below it, there are two tabs: 'User Signup' (active) and 'NGO Signup'. The 'User Signup' section contains a title 'User Signup', followed by input fields for Email, Password, Phone, Address, Name, and Age. At the bottom, there is a yellow 'Signup' button and a link: 'Already have an account? [Login](#)'.

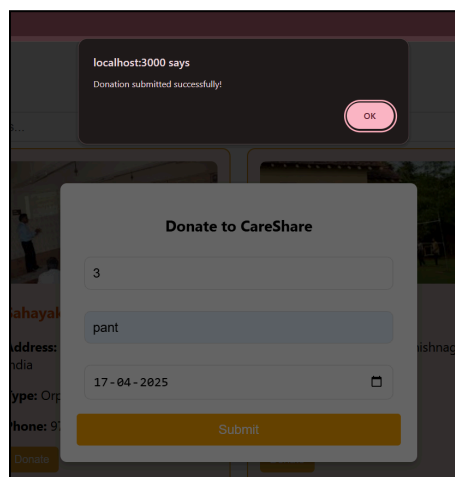
Home page -



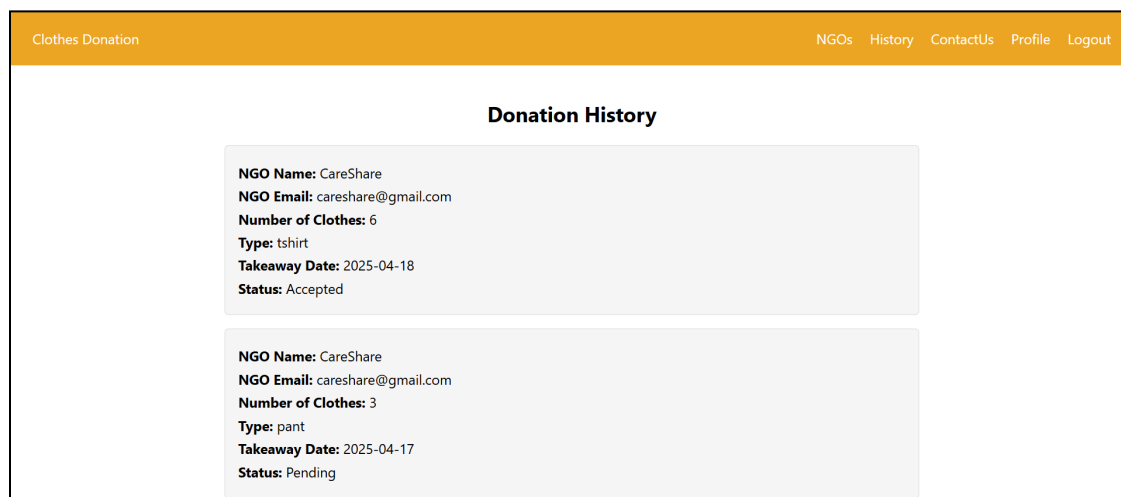
Ngos listing page -



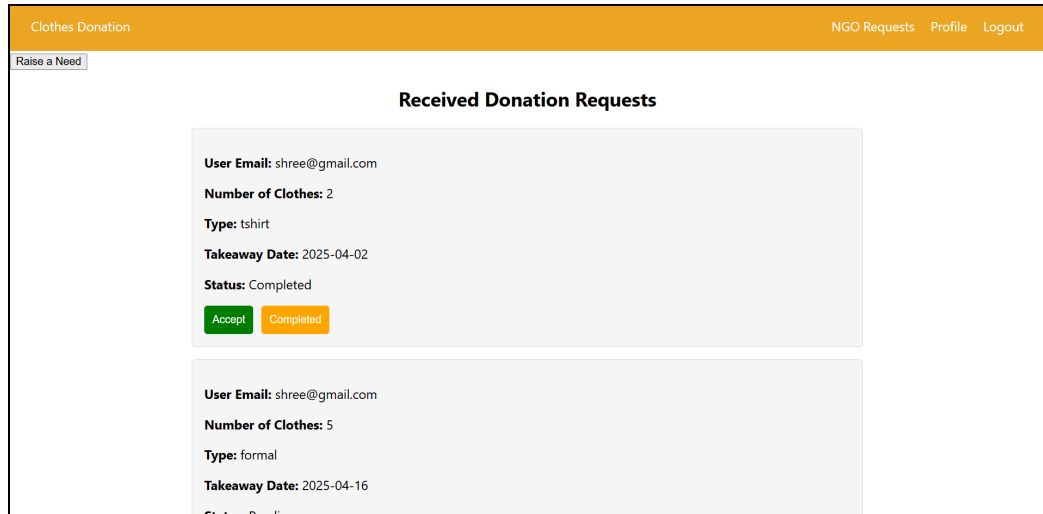
Donation Request -



Donation History of user -

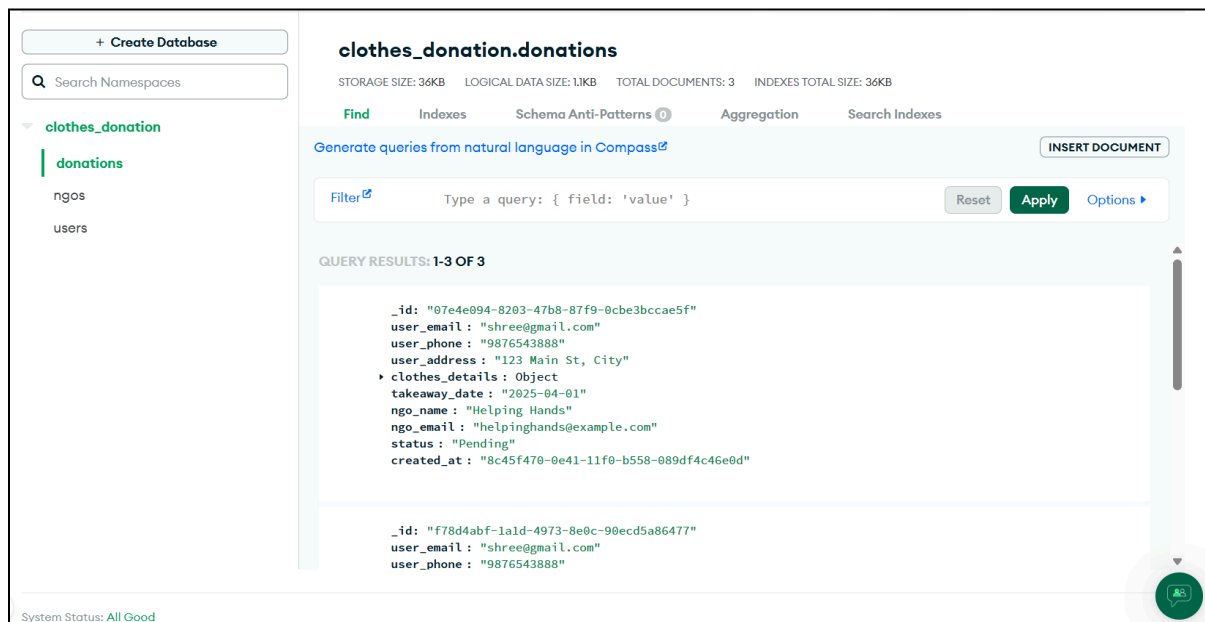


Ngos Page of requests from users -

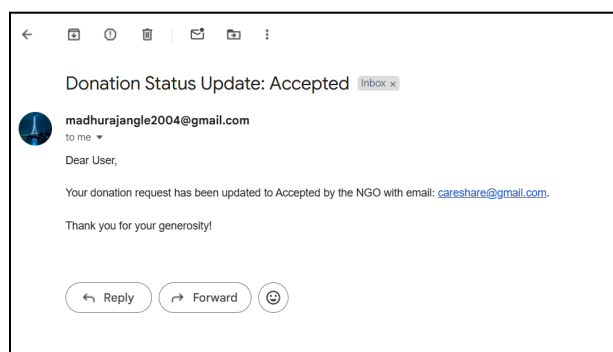


4.2 Output (Database)

MongoDB collection -

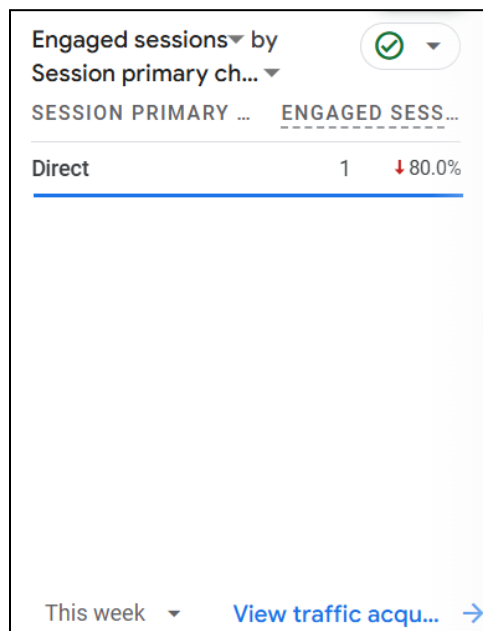
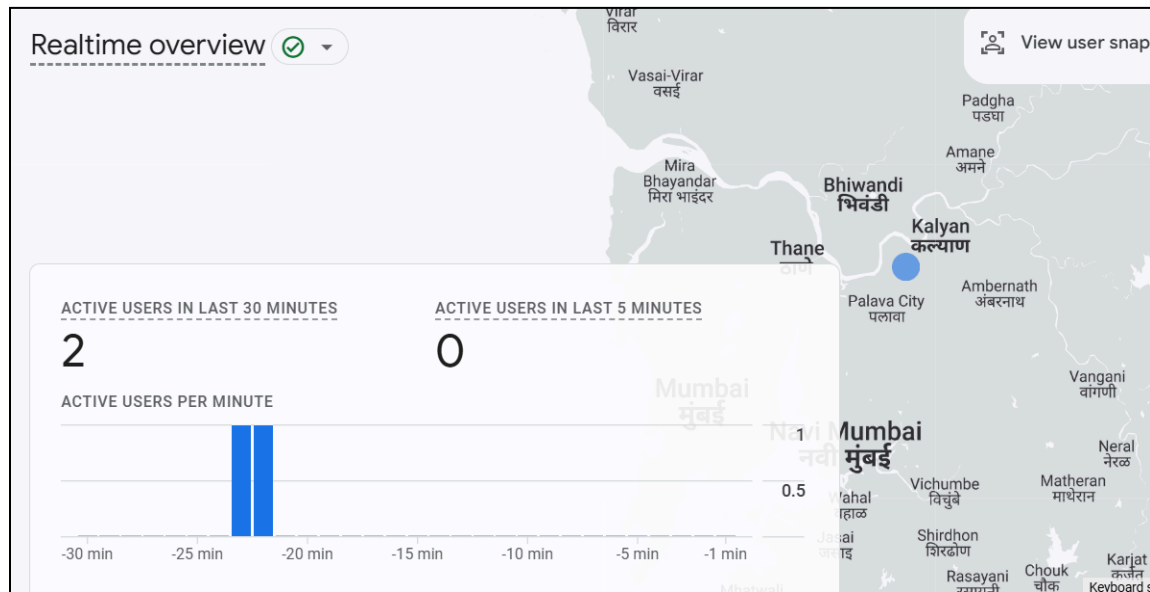
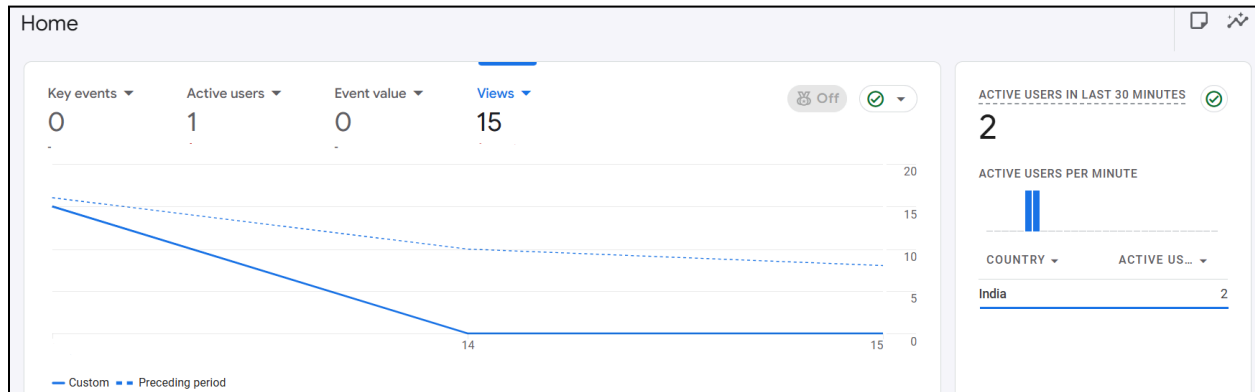


Email Functionality-



Chapter 5 - Results

5.1 Google Analytics



5.2 Conclusion

The Clothes Donation App offers a simple and efficient way for users to donate clothes to NGOs. Built with React, Flask, and MongoDB, the app ensures smooth communication between donors and NGOs with features like request management, donation tracking, and email notifications. With proper setup, the application provides a reliable and user-friendly platform to support meaningful contributions to society.

5.3 Future Scope

- Mobile App Development: Creating Android/iOS versions to increase accessibility and reach more users.
- AI-Based Suggestions: Implementing AI to recommend NGOs based on user location, donation type, and history.
- Image Recognition: Adding image upload and categorization features for better sorting of donated clothes.
- Multi-Item Expansion: Extending the platform to include donations of shoes, books, and other essentials.

References

1. <https://ieeexplore.ieee.org/document/10059757>
2. <https://ieeexplore.ieee.org/document/10864148>
3. <https://ieeexplore.ieee.org/document/9375945>